GOOGLE DATA ANALYTICS CAPSTONE · CYCLISTIC BIKE SHARE CASE STUDY

How Do Riders Use Cyclistic Bikes Differently?

Peter Francis Muthukkalai

5,800+

bikes across
692 stations
in Chicago

CYCLISTIC FLEET

BACKGROUND

The Business Task

"How do annual members and casual riders use Cyclistic bikes differently?"
— Lily Moreno, Director of Marketing

Why it matters

Finance analysts confirm annual members are significantly more profitable than casual riders.

The opportunity

Casual riders already know Cyclistic. Converting them—not finding new customers—is the priority.

Analyst's scope

Identify behavioural differences to inform a targeted conversion campaign.

DATA & PROCESS

Prepare, Process & Tools

DATA SOURCE

Cyclistic Historical Trip Data (12 months)

— Provided by Motivate International Inc. under public licence

— Covers ride start/end times, station locations, bike type and rider type

— No personally identifiable information — privacy preserved

— ~5.7 million rows of ride records across 12 CSV files

TOOLS USED

Microsoft Excel

Primary analysis environment — pivot tables and charting

Power Query

Data transformation — combined 12 CSV files, removed nulls and duplicates, standardised column types

Pivot tables

Aggregated ride counts and average durations by rider type, day, month and hour

DATA & PROCESS

Data Cleaning — Assumptions & Decisions

Before analysis, the dataset was audited for quality issues. The following records were excluded based on clearly defined assumptions:

Negative Ride Length

EXCLUDED

A ride end time earlier than its start time is physically impossible. These records indicate data entry errors or system clock issues and were removed entirely.

Ride Length > 24 hours and < 2 hours

EXCLUDED

Outliers at both ends of the duration range were excluded: rides over 24 hours suggest an system or clocking error, while rides under 2 minutes could be due to equipment faults.

Blank rows

EXCLUDED

Rows missing start/end time cannot be reliably used in aggregations and were dropped during Power Query ingestion.

Duplicate records

EXCLUDED

Exact duplicate rows — identical ride ID were identified and removed to avoid double-counting rides in pivot table analysis.

THE CORE FINDING

Two very different riders, one bike network.

CASUAL RIDER

Rides for leisure & recreation

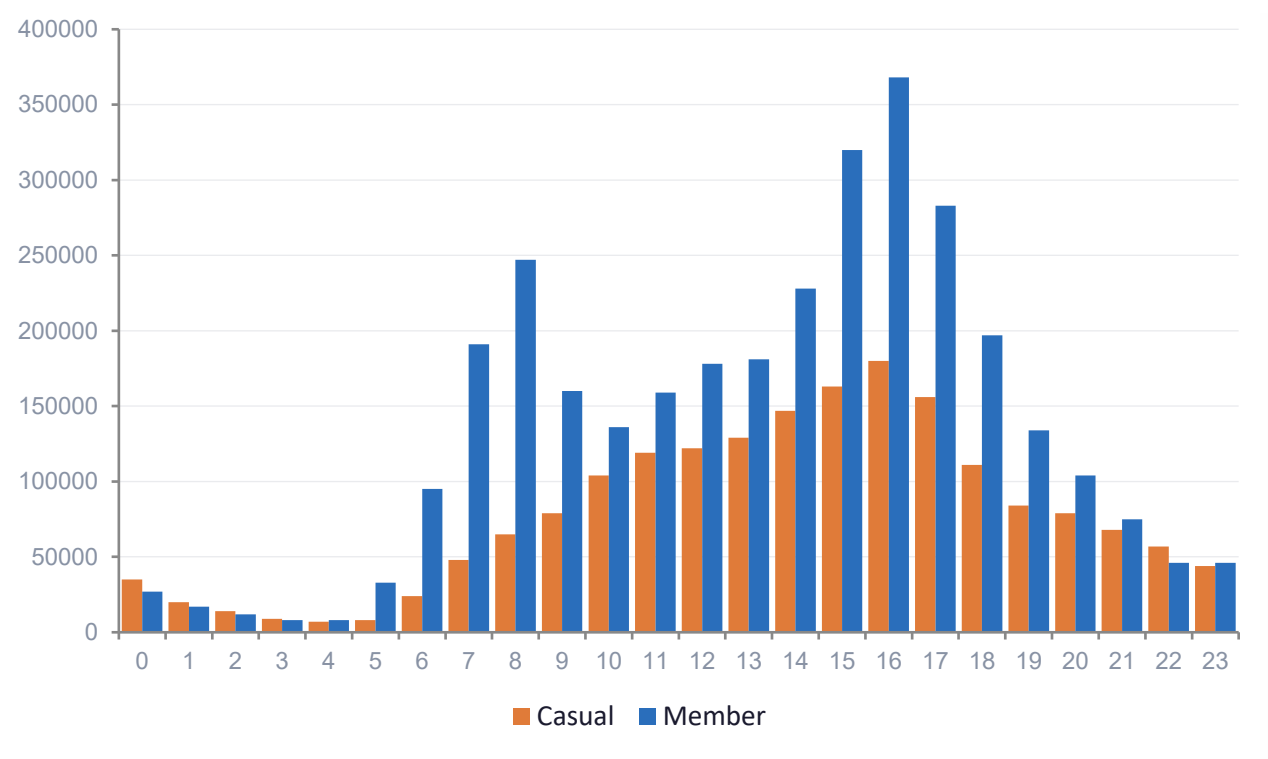
- Peaks Saturday & Sunday afternoons
- ~20 min average ride — longer, exploratory trips
- Concentrated in summer months (May–Sep)
- Gradual mid-morning build, no commute spikes
- More 30–60 min rides than members

ANNUAL MEMBER

Rides to commute to work

- Sharp spikes at 8am and 5pm on weekdays
- ~12.5 min average ride — direct, purposeful trips
- Consistent usage year-round, even in winter
- Monday–Friday dominates; dip on weekends
- Overwhelmingly short rides under 15 minutes

Riding by Hour — The Commute Signal



What the chart shows

8am peak

Members surge at 8am — the morning commute. A typical member is cycling to the office.

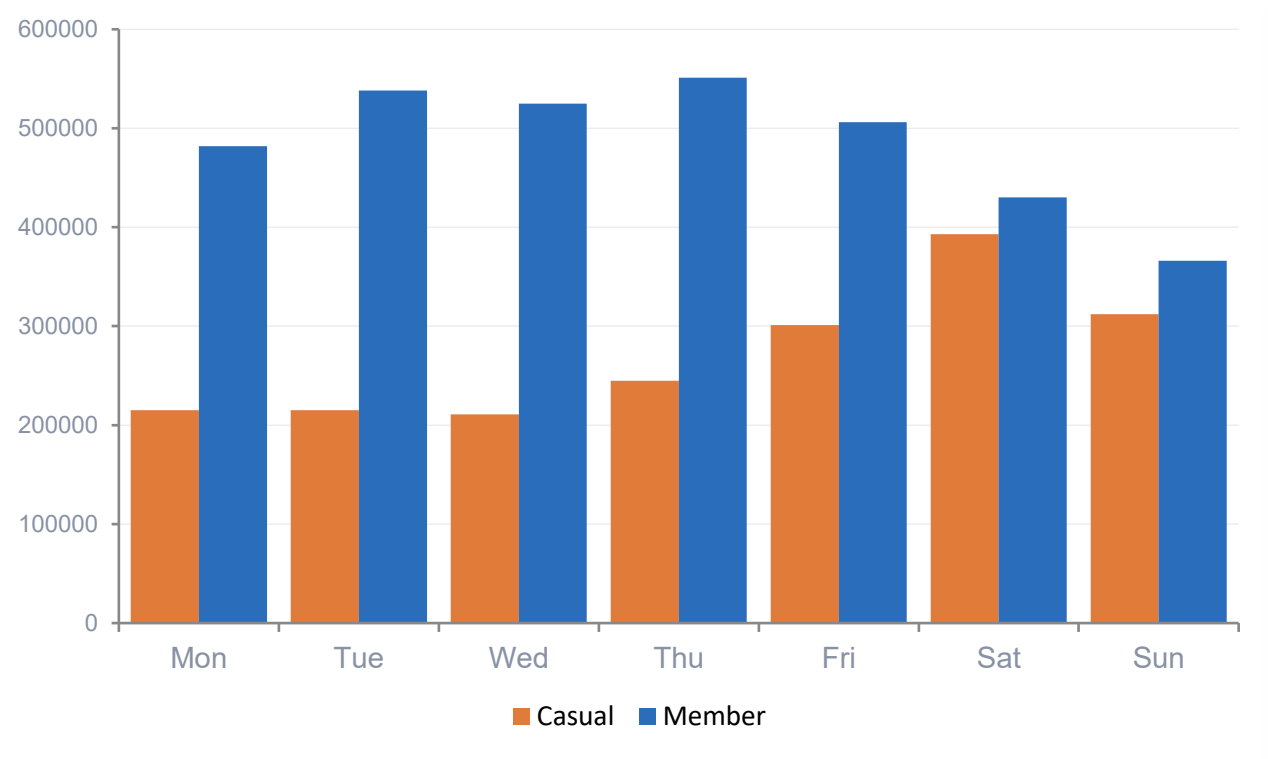
5pm peak

Second member spike at 5pm confirms the return commute pattern.

Mid-day curve

Casuals build steadily from 10am, plateau through the afternoon — a leisure riding profile.

Riding by Day — Weekday vs Weekend



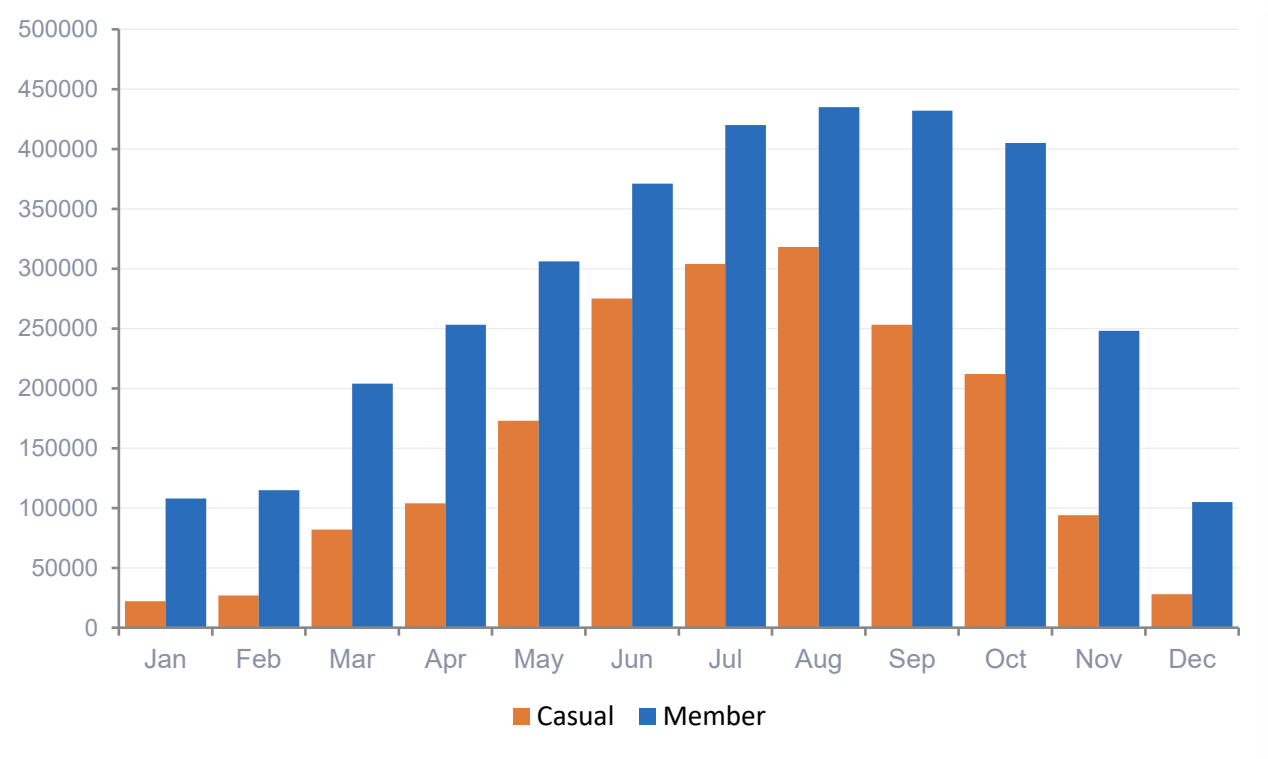
What the chart shows

Saturday peak
Casuals hit ~393K rides on Saturday — nearly double their weekday average. Weekend recreation.

Weekday dominance
Members are remarkably stable Mon–Fri. This is habitual, utility-driven behaviour.

Weekend dip
Members actually ride less on weekends — confirming the commute-first use case.

Riding by Month — The Seasonal Gap



What the chart shows

Summer window

Casual rides peak June–Aug at 300K+. This is the prime conversion opportunity window.

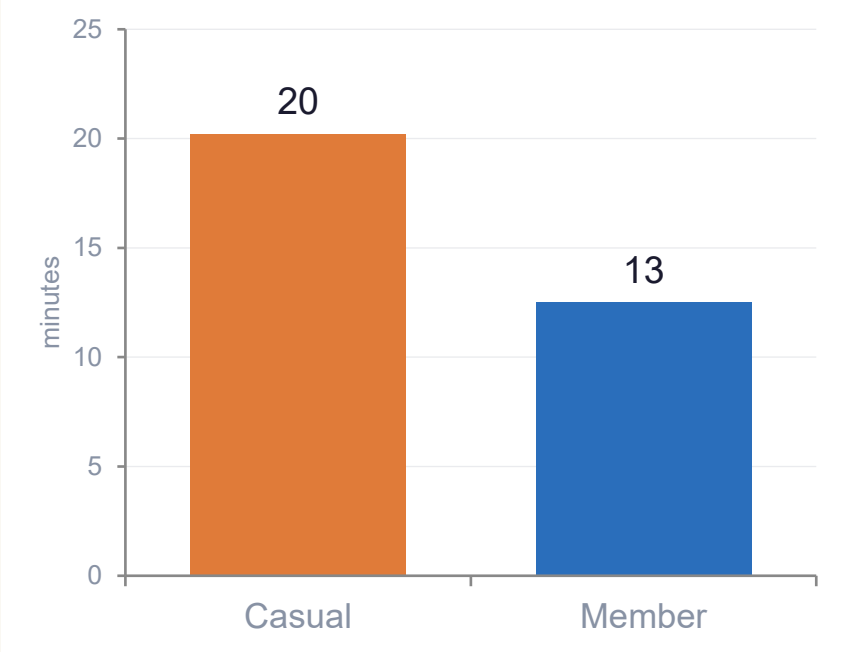
Winter cliff

Casual rides drop ~93% by January — highly weather-dependent, discretionary use.

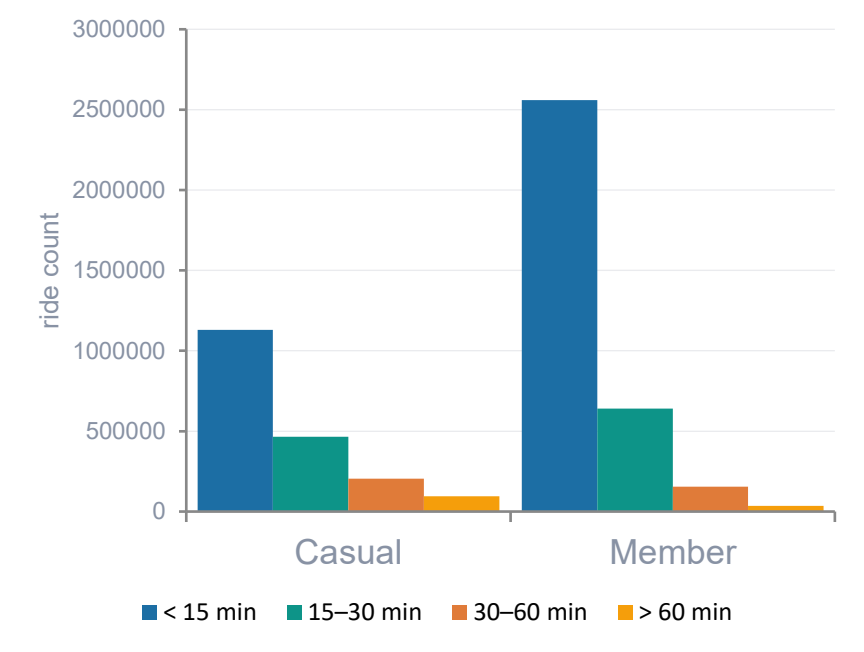
Year-round habit

Members maintain 25%+ of peak volume in winter — bikes are a genuine transport tool for them.

Ride Duration — Leisure vs Purpose



Average ride length



Rides by duration bracket

Casuals ride 62% longer on average, with more 30–60 min trips — consistent with exploration and leisure, not commuting.

RECOMMENDATIONS

Three Strategies to Convert Casual Riders

01 · HIGHEST PRIORITY

Summer Weekend Conversion Campaign

Run targeted digital and in-app campaigns on Fridays–Sundays between May and August — when casual ridership peaks at ~393K rides/Saturday. Offer a time-limited annual membership discount at high-traffic leisure stations (parks, waterfronts, tourist areas). Messaging: 'You already ride with us — make it year-round.'

02 · MEDIUM PRIORITY

Discounted Membership Trial for Commute-Adjacent Riders

Identify casual riders who use Cyclistic on 3+ weekdays in a month — they are already commute-adjacent. Offer a discounted 30-day membership trial with push notifications timed to the 8am and 5pm commute peaks. Lower the financial barrier to entry and frame it around a real use case: 'Cycle to work — your morning commute sorted.'

03 · RETENTION & PIPELINE

Winter Bridge Pass + Early-Bird Membership

Note: Outside primary scope, but warrants consideration. Introduce a 3-day weekend pass (Fri–Sun) as a stepping-stone product in off-peak months (Nov–Feb). Casual rides drop 93% in winter — a low-cost pass maintains engagement and brand touchpoints. Pair with an early-bird annual membership offer redeemable the following spring to convert engaged casuals before peak season.

NEXT STEPS

Applying These Insights

Validate with qualitative data

Conduct short rider surveys at high-casual stations to test whether commuting is a stated barrier — and what pricing would motivate conversion.

Geo-analysis of top casual stations

Identify which docking stations over-index for casual weekend traffic to focus physical marketing (signage, QR codes, promotions) in the right places.

A/B test the summer campaign

Run two creative variants in July: one price-led offer, one lifestyle/commute-framing message. Measure conversion rate and track through Q4.

Track trial-to-member conversion rate

Measure what percentage of discounted trial members convert to full annual membership after 30 days — the single most important metric for the campaign.

The data is clear.

Annual members rely on Cyclistic to get to work.
Casual riders use it to enjoy the city.

Converting casual riders means showing them that the bike they enjoy on Saturday can also be the one that gets them to work on Monday. Summer is the conversion window.

Thank You

Peter Francis Muthukkalai

Rider Behaviour Analysis

Google Data Analytics Capstone · Cyclistic Bike-Share